Tanmay Bhuskute

+1 (732) 858-3027 | tanmay.v.bhuskute@gmail.com | github.com/Tanmayb05 | linkedin.com/in/tanmay-bhuskute
Seattle, WA

SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production engineering experience building fault-tolerant distributed infrastructure and profiling systems at scale using Python, Java, and AWS. Current research applies PyTorch-based pipelines and end-to-end performance profiling to isolate compute and inference bottlenecks in large model workloads.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, C/C++, SQL, Go, JavaScript, TypeScript.

ML & Distributed Training: PyTorch, TensorFlow, scikit-learn, LangChain, parallelism concepts, end-to-end performance profiling, latency/throughput benchmarking.

Infrastructure: AWS, Docker, Kubernetes, ArgoCD, Helm, Terraform, Linux, Bash, fault-tolerant large-scale systems.

Backend & Data: REST APIs, PostgreSQL, MySQL, pandas, NumPy, CI/CD.

Developer Tools: Git, GitLab CI/CD, Jenkins, SonarQube, Azure, Google Cloud Platform.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Built PyTorch-based evaluation pipeline profiling model runs end to end across a two-stage architecture to isolate whether latency was bound by inference, memory, or per-step overhead, benchmarked against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Traced a redundant per-step inference bottleneck to a single stage of the pipeline and drove the fix there, demonstrating **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, profiling and diagnosing live infrastructure issues across Linux, EC2, databases, and Teamcenter X services.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%**.
- Migrated **40+ production environments** from process-based deployments to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern —- Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [PySpark, Delta Lake, scikit-learn, FastAPI] Jul 2026

- Built distributed ML pipeline forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** hourly dataset) on Apache Spark; designed two-stage Poisson hurdle model improving MAPE **18.77% to 16.52%** and cutting RMSE **43%** over baseline Random Forest.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).